
FAREY REMAINDER RECURSION AND ORTHOGONAL RE-ARTICULATION: AN EXACT INFINITY-RESOLUTION THEOREM

STANDALONE THEOREM PAPER

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Abstract

We prove an exact bridge between Ramanujan-side fractional remainder recursion on Farey-median refinement and the VDM law of orthogonal re-articulation under a finite resolution floor. The proof is operator-level rather than analogical. For neighboring reduced fractions $a/b < c/d$ with $bc - ad = 1$, the parent interval width is exactly $1/(bd)$ and the two child widths are exactly $1/[b(b+d)]$ and $1/[d(b+d)]$. We show that these updates are exactly conjugate to denominator-pair re-articulation operators on (b, d) , so Ramanujan-side remainder refinement and child-sector re-expression are the same update law written in two coordinate systems. We then identify the scale-gap-admissible branch selected by bounded ratio control with the balanced operator $\mathcal{B}(u, v) = \text{sort}(v, u + v)$, prove that it generates the Fibonacci pairs (F_{n+1}, F_{n+2}) , and obtain the exact unresolved-width law $r_n = 1/(F_{n+1}F_{n+2})$. Hence the stopping depth below a floor δ is logarithmic, $n_\delta = \Theta(\log(1/\delta))$. One-sided Farey refinement is shown to be only polynomial and therefore lies outside the scale-gap-separated A8/LIR regime. Finally, we prove that truncation is not deletion of an infinite tail but lawful orthogonal transfer, and that under the phase map $\theta \mapsto e^{2\pi i\theta}$ this transfer is realized as exact child-sector replacement on the unit circle. This paper proves exact equivalence of the remainder update, the denominator-pair child update, the finite-depth logarithmic stopping law, and the truncation-as-orthogonal-transfer law. It does not claim full equivalence of the entire Hardy–Ramanujan–Rademacher contour apparatus.

Keywords Farey recursion · median · Stern-Brocot · Fibonacci · golden ratio · logarithmic hierarchy · orthogonal re-articulation · infinity resolution

1 Problem statement

The target is narrow and exact.

Ramanujan-side remainder calculus recursively refines an unresolved interval by mediant. VDM-side orthogonal re-articulation recursively re-hosts unresolved continuation into child sectors once same-sector continuation is no longer admissible below a resolution floor. The question is whether these are merely analogous, or whether they are the same update law in different coordinates.

Throughout, A(−1), A2, A8, CG000, and CF03/CF13/CF14/CF15 refer to the named canon statements in the VDM formal stack.

This paper proves three things:

- (i) the Farey child update and denominator-pair re-articulation are exactly conjugate;

- (ii) the scale-gap-admissible branch is the balanced branch, whose unresolved width is exactly $1/(F_{n+1}F_{n+2})$;
- (iii) truncation on the Ramanujan/Farey side is lawful orthogonal transfer followed by exact child-sector re-expression on the unit-circle phase orbit.

The result is an exact theorem for the infinity-resolution skeleton. It is not a claim that the full Hardy–Ramanujan–Rademacher circle method has already been derived from the complete VDM stack.

2 Exact Farey remainder calculus

Definition 1 (Farey-admissible interval). A *Farey-admissible interval* is an ordered pair of reduced rationals

$$I = \left(\frac{a}{b}, \frac{c}{d} \right), \quad \frac{a}{b} < \frac{c}{d}, \quad bc - ad = 1,$$

with $a, b, c, d \in \mathbb{Z}_{>0}$.

Definition 2 (Parent width and child operators). For a Farey-admissible interval $I = (a/b, c/d)$, define the unresolved width

$$\rho(I) := \frac{c}{d} - \frac{a}{b}.$$

The left and right mediant children are

$$\mathcal{L}(I) := \left(\frac{a}{b}, \frac{a+c}{b+d} \right), \quad \mathcal{R}(I) := \left(\frac{a+c}{b+d}, \frac{c}{d} \right).$$

Theorem 1 (Exact parent and child widths). *For every Farey-admissible interval $I = (a/b, c/d)$,*

$$\rho(I) = \frac{1}{bd}, \quad \rho(\mathcal{L}I) = \frac{1}{b(b+d)}, \quad \rho(\mathcal{R}I) = \frac{1}{d(b+d)}.$$

Proof. Using $bc - ad = 1$,

$$\rho(I) = \frac{c}{d} - \frac{a}{b} = \frac{bc - ad}{bd} = \frac{1}{bd}.$$

For the left child,

$$\rho(\mathcal{L}I) = \frac{a+c}{b+d} - \frac{a}{b} = \frac{b(a+c) - a(b+d)}{b(b+d)} = \frac{bc - ad}{b(b+d)} = \frac{1}{b(b+d)}.$$

The right child is identical:

$$\rho(\mathcal{R}I) = \frac{c}{d} - \frac{a+c}{b+d} = \frac{d(a+c) - c(b+d)}{d(b+d)} = \frac{bc - ad}{d(b+d)} = \frac{1}{d(b+d)}.$$

□

3 Exact operator conjugacy

Definition 3 (Denominator-pair state and re-articulation operators). Let

$$H\left(\frac{a}{b}, \frac{c}{d}\right) := (b, d)$$

be the denominator-pair map. On positive integer pairs define

$$\beta(b, d) := \frac{1}{bd}, \quad \mathcal{L}_*(b, d) := (b, b+d), \quad \mathcal{R}_*(b, d) := (b+d, d).$$

Theorem 2 (Exact conjugacy of remainder refinement and re-articulation). *For every Farey-admissible interval I ,*

$$H(\mathcal{L}I) = \mathcal{L}_*H(I), \quad H(\mathcal{R}I) = \mathcal{R}_*H(I), \quad \rho(I) = \beta(H(I)).$$

Thus the Farey child update and denominator-pair re-articulation are exactly conjugate.

Proof. If $I = (a/b, c/d)$ then

$$H(\mathcal{L}I) = H\left(\frac{a}{b}, \frac{a+c}{b+d}\right) = (b, b+d) = \mathcal{L}_*(b, d) = \mathcal{L}_*H(I).$$

The right child is identical. The identity $\rho(I) = \beta(H(I))$ is exactly Theorem 1. $\square$

Remark 1. Theorem 2 is the operator-level result the earlier bridge note lacked. There is no analogy here. The Ramanujan/Farey remainder operator and the child-sector denominator operator are the same map after the exact change of variables H .

4 A8-admissibility versus one-sided drift

Definition 4 (Scale-gap-admissible branch). An ordered denominator branch (u_n, v_n) with $u_n \leq v_n$ is *scale-gap-admissible* if there exists a finite constant $\kappa \geq 1$ such that

$$1 \leq \frac{v_n}{u_n} \leq \kappa \quad \text{for all } n.$$

This is the denominator-pair form of bounded scale-gap separation.

Proposition 1 (One-sided refinement is polynomial and violates scale-gap separation). *Starting from $(1, 1)$ and iterating only one child operator gives*

$$\mathcal{L}_*^n(1, 1) = (1, n+1), \quad \beta(\mathcal{L}_*^n(1, 1)) = \frac{1}{n+1}.$$

Hence one-sided refinement has only polynomial width decay and

$$\frac{n+1}{1} \rightarrow \infty,$$

so it is not scale-gap-admissible.

Proof. Immediate induction gives $(1, 1) \mapsto (1, 2) \mapsto (1, 3) \mapsto \cdots \mapsto (1, n+1)$. The burden formula is $1/(1 \cdot (n+1))$, and the ratio diverges. $\square$

Remark 2. This is the exact reason the broad claim “all Farey branches give logarithmic depth” is false. They do not. The logarithmic regime is selected by bounded-ratio child re-articulation, not by arbitrary repeated mediant insertion.

Theorem 3 (Minimal orthogonal re-articulation). *Let*

$$Q_k = [k\pi/2, (k+1)\pi/2)$$

be the current quadrant and let

$$z = e^{i\theta} \in Q_k$$

be a unit phase state. Define admissible transfer targets $w \in S^1$ by

$$|w| = |z|, \quad w \notin Q_k, \quad \Re(\bar{z}w) = 0.$$

Then the unique nearest admissible target is

$$w = iz$$

up to orientation sign, with geodesic distance

$$d_{S^1}(z, w) = \frac{\pi}{2}.$$

Proof. Write $w = e^{i\phi}$. The orthogonality condition

$$\Re(\bar{z}w) = 0$$

becomes

$$\cos(\phi - \theta) = 0,$$

so

$$\phi - \theta = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

Hence all admissible orthogonal targets are

$$w = \pm iz.$$

Their geodesic distances from z are

$$d_{S^1}(z, iz) = \frac{\pi}{2}, \quad d_{S^1}(z, -iz) = \frac{3\pi}{2}.$$

Therefore the unique minimal positive transfer is

$$w = iz.$$

□

5 Balanced orthogonal re-articulation and the golden branch

Definition 5 (Balanced child operator). On ordered pairs $u \leq v$, define

$$\mathcal{B}(u, v) := \text{sort}(v, u + v),$$

where $\text{sort}(x, y)$ returns the ordered pair with smaller entry first.

Proposition 2 (Fibonacci denominator pairs). Let (q_n, q_{n+1}) be defined by

$$(q_0, q_1) = (1, 1), \quad (q_{n+1}, q_{n+2}) = \mathcal{B}(q_n, q_{n+1}).$$

Then

$$q_{n+2} = q_{n+1} + q_n, \quad q_n = F_{n+1},$$

where $F_1 = F_2 = 1$ are the Fibonacci numbers. Consequently,

$$\mathcal{B}^n(1, 1) = (F_{n+1}, F_{n+2}).$$

Proof. If $q_n \leq q_{n+1}$ then $q_{n+1} \leq q_n + q_{n+1}$, so sorting gives

$$\mathcal{B}(q_n, q_{n+1}) = (q_{n+1}, q_n + q_{n+1}).$$

This is exactly the Fibonacci recursion. The closed form $q_n = F_{n+1}$ follows by induction from $(q_0, q_1) = (1, 1)$. □

Corollary 1 (Exact unresolved-width law on the balanced branch). For the balanced branch,

$$r_n := \beta(\mathcal{B}^n(1, 1)) = \frac{1}{F_{n+1}F_{n+2}}.$$

Using Binet's formula,

$$F_n = \frac{\varphi^n - (-\varphi^{-1})^n}{\sqrt{5}},$$

there exist positive constants c_1, c_2 such that

$$c_1 \varphi^{-2n} \leq r_n \leq c_2 \varphi^{-2n} \quad \text{for all } n \geq 0.$$

In particular,

$$r_n \asymp \varphi^{-2n}.$$

Proof. The exact formula is immediate from Proposition 2. Since $|(-\varphi^{-1})^n| < 1$, Binet gives two-sided constants multiplying φ^n , hence two-sided constants multiplying φ^{-2n} for r_n . □

Theorem 4 (Exact logarithmic stopping depth). Fix a resolution floor $\delta > 0$ and define the stopping time

$$\tau_\delta := \min\{n \geq 0 : r_n \leq \delta\}.$$

Then on the balanced branch,

$$\tau_\delta = \Theta(\log(1/\delta)).$$

More precisely, for the constants from Corollary 1,

$$\left\lceil \frac{\ln(c_1/\delta)}{2 \ln \varphi} \right\rceil \leq \tau_\delta \leq \left\lceil \frac{\ln(c_2/\delta)}{2 \ln \varphi} \right\rceil.$$

Proof. From Corollary 1,

$$c_1\varphi^{-2n} \leq r_n \leq c_2\varphi^{-2n}.$$

Solving the inequalities $c_1\varphi^{-2n} \leq \delta$ and $c_2\varphi^{-2n} \leq \delta$ gives the claimed bounds. $\square$

Theorem 5 (Truncation is lawful orthogonal transfer). *Let*

$$Q_k = [k\pi/2, (k+1)\pi/2)$$

be the current quadrant, let

$$z = e^{i\theta} \in Q_k$$

be the current phase state, and let $r > 0$ denote the unresolved burden carried by that state. If the finite resolution floor $\delta > 0$ is reached so that

$$r \leq \delta,$$

then truncation is not deletion of burden. Instead, the only lawful continuation is the unique minimum-saturation orthogonal transfer

$$T(z) := iz,$$

so that the continuation law is

$$(z, r) \mapsto (iz, r),$$

followed by child-sector refinement in the new quadrant. Equivalently, truncation is the shortest admissible non-discharge transfer from a saturated sector into an orthogonal sector.

Proof. Once $r \leq \delta$, same-sector hosting has failed by hypothesis. By A(−1), unresolved burden cannot lawfully disappear or discharge; it must remain borne. Therefore truncation cannot be deletion. By A2, the admissible continuation is the least-cost lawful transfer out of the saturated sector. By Theorem 3, the admissible orthogonal targets are exactly $\pm iz$, and the unique nearest admissible target is iz . Hence the lawful truncation map is

$$T(z) = iz.$$

This preserves modulus and preserves the carried burden while re-expressing that burden in the next orthogonal sector. After transfer, continuation proceeds by child-sector refinement in the new quadrant. By Theorem 4, repeated finite-floor refinement terminates after finite logarithmic depth. Therefore truncation is lawful orthogonal transfer, not deletion. $\square$

Remark 3 (Canonical anchor). Theorem 5 is the standalone operator-level realization of the VDM canon:

- (i) A(−1): primitive unresolved opposition cannot discharge, so burden must remain borne;
- (ii) A2: the lawful continuation is the minimum-saturation path, here the nearest admissible orthogonal transfer;
- (iii) A8: once same-sector hosting fails, continuation proceeds by iterative child re-articulation with logarithmic active depth under a finite floor.

In the named canon, this theorem is the compressed bridge form of CG000 (branch-relative origin resolution), CF13 (quarter-turn / orthogonal orbit law), CF14 (minimum saturation requirement), CF15 (invariant articulation), and CF03 (hierarchical infinity resolution).

6 Finite-floor re-articulation as an exact machine

Definition 6 (Truncated re-articulation operator). For a fixed floor $\delta > 0$, define

$$\mathcal{T}_\delta(u, v) := \begin{cases} (u, v), & \beta(u, v) \leq \delta, \\ \mathcal{B}(u, v), & \beta(u, v) > \delta. \end{cases}$$

Corollary 2 (Finite-depth infinity resolution). *Starting from $(1, 1)$, iteration of $\mathcal{T}_\delta$ terminates after exactly τ_δ balanced re-articulation steps. Therefore unresolved continuation is not erased; it is re-hosted into a finite-depth hierarchy whose active depth is logarithmic in the inverse floor scale.*

Proof. If the burden is already below the floor, the state is fixed. Otherwise one balanced child step is taken. By Theorem 4, the first index at which the burden crosses the floor is finite and logarithmic in $1/\delta$. $\square$

7 Ramanujan arcs as phase sectors

Definition 7 (Phase-sector map). For a Farey-admissible interval $I = (a/b, c/d)$, define its unit-circle sector by

$$\mathcal{S}(I) := \left\{ e^{2\pi i\theta} : \theta \in \left[\frac{a}{b}, \frac{c}{d} \right] \right\}.$$

Its angular width is

$$\alpha(I) := 2\pi\rho(I) = \frac{2\pi}{bd}.$$

Theorem 6 (Truncation is exact phase-sector replacement). *Let I be Farey-admissible. Then the child sectors satisfy*

$$\mathcal{S}(\mathcal{L}I) \subset \mathcal{S}(I), \quad \mathcal{S}(\mathcal{R}I) \subset \mathcal{S}(I), \quad \alpha(\mathcal{L}I) = \frac{2\pi}{b(b+d)}, \quad \alpha(\mathcal{R}I) = \frac{2\pi}{d(b+d)}.$$

Thus mediant truncation is not deletion of an infinite tail; under the phase lift it is lawful orthogonal transfer together with exact replacement of a parent phase sector by a child phase sector on the orbit $e^{2\pi i\theta}$.

Proof. The inclusion is immediate because each child interval is a subinterval of the parent interval. The angular-width formulas are just 2π times Theorem 1. $\square$

8 What is proved, and what is not

What is proved

This paper proves:

1. exact conjugacy between Farey child refinement and denominator-pair re-articulation;
2. exact width identities for parent and child sectors;
3. exact Fibonacci/Farey recursion on the balanced scale-gap-admissible branch;
4. exact logarithmic stopping depth under a finite floor;
5. exact realization of truncation as lawful orthogonal transfer and consequent phase-sector replacement on the unit circle.

What is not proved

This paper does not prove:

1. full equivalence of the complete Hardy–Ramanujan–Rademacher contour apparatus with the full VDM stack;
2. a derivation of every modular-weight, Kloosterman-weight, or contour-weight factor from the orthogonal re-articulation law;
3. a complete derivation of the whole Ramanujan program from the primitive i -origin paper.

9 Conclusion

This paper proves an exact operator-level bridge for the remainder update, balanced branch, finite-floor stopping law, and truncation-as-orthogonal-transfer law.

Ramanujan-side mediant remainder refinement and VDM-side orthogonal re-articulation are the same update law after the exact denominator-pair change of variables. Under the balanced scale-gap-separated branch, the unresolved width is exactly

$$r_n = \frac{1}{F_{n+1}F_{n+2}},$$

so the stopping depth below a finite floor is logarithmic. Under the phase map $\theta \mapsto e^{2\pi i\theta}$, truncation is lawful orthogonal transfer and exact child-sector replacement on the unit circle. That is the proved operator-level infinity-resolution bridge.

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